University of Pretoria

SEMESTER TEST: 13 MARCH 2023

COURSE: Computer Science
PAPER: COS332

TIME: 1 hour
MARKS: 40

12

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED
YOU ARE ALLOWED TO WRITE ON THIS TEST PAPER

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) A proprietary network protocol is a protocol that

A: Has been standardised by some national standards body.
B: Is owned by a specific vendor.
C: Is developed as an open source protocol.
D: Is now so old that using it has been deprecated.
E: Describes the properties of other network protocols.

1 b) Which of the following characters is not present in standard ASCII?

American Standard

A: A
B: &
C: #
D: £ (thus no need for the £ sign)
E: ~

7.
6.
5.
4.
3.

COURSE: Computer Science
PAPER: COS332

c) Suppose node X communicates with node Z. A header is added at layer n at X. This header may be removed by

A: Layer n at Z.
B: Layer $n-1$ at X.
C: Layer $n+1$ at X.
D: Layer $n-1$ at Z.
E: More than one of the above

peer to peer communication, thus must be layer n.
usually header is used by dest node and then discarded.

d) Which ISO layer enables process-to-process communication via the network?

i.e. a pipe for communication.

A: 6
B: 5 Session
C: 4 Transport
D: 3 Network
E: 2

e) A layer on a protocol stack communicates with its peer layer. Suppose node X sends a message to node Y. Which node's (or nodes') layer 3 is/are deemed to be the peer(s) of layer 3 of X?

A: Only Y.
B: All the nodes on the path after X, up to and including Y.
C: All the nodes on the path from X (including X), up to and including Y.
D: Only the router nodes on the path after X, up to and including Y, which excludes most nodes on this path.
E: Only the router nodes on the path from X (including X), up to and including Y, which excludes most nodes on this path.

f) On which OSI layer(s) is error checking primarily performed?

A: 7

B: 2

C: Application-oriented layers

D: Network-oriented layers

E: All seven layers   each layer has its own form of error checking

g) The OSI layer that, amongst others, attempts to emulate database transactions where messages can be 'rolled back' if the final message in a sequence is not sent, is layer

A: 2 Data Link

B: 3 Network

C: 4 transport

D: 5 session

E: 6 Presentation

h) Which protocol is primarily intended to manipulate a mailbox on a server?

A: IMAP4

B: POP3   retrieve/receive email

C: SMTP   send email

D: More than one of the above

E: All of the above

i) The capetown. sTLD has to adhere to policies prescribed by

A: ICANN

B: ZADNA

C: ISO

D: More than one of the above

E: All of the above

SFTP < 25 587

j) The SMTP submit port is

A: 25

B: 100

C: 443

D: 587

E: None of the above

k) With which architecture was the *Napster* service associated?

A: Client-server

B: Peer-to-peer

C: 3-tier

D: 4-tier

E: 5-tier

l) The popular network software known as `bind`

A: Acts as a name server.

B: Connects two processes to facilitate communication.

C: Associates a process with a port.

D: Is used to authenticate a user.

E: More than one of the above

m) When a server opens a socket for communication with a remote host, it needs to supply

A: The port number on the remote host.

B: The address of the remote host.

C: The port number on the local host.

D: More than one of the above

E: All of the above

5 1 a 2 b 3 c 4 d 5 e 6 f 7 g 8 h 9 i 10 j 11 k 12 l 13 m

n) If the FQDNs of DNS root name servers are ordered lexicographically, the last entry in the list is

A: `l.dns.net`

B: `m.dns.net`

C: `n.dns.net`

D: `l.root-servers.net`

E: `m.root-servers.net`

o) An agent that monitors and controls network activity in an SNMP-managed network node, is a a(n)

A: Client

B: Server

C: P2P-node

D: More than one of the above

E: All of the above

p) Which condition code will be returned by an SMTP server after successfully handling a `helo` or `ehlo` message?

A: 0

B: 150

C: 250

D: 350

E: 450

q) The bulk, if not all, of the resource records in the DNS root name servers are of the following type(s):

A: A

B: MX

C: NS

D: More than one of the above

E: All of the above

r) Which of the following is a popular application used to inspect traffic that flows on a network?

A: SniffAndSnort

B: `telnet`

C: `nslookup`

D: `tracert / traceroute`

E: Wireshark

s) Which South African Act specifically regulates 'wiretapping'?

A: POPIA

B: RICA

C: PAIA

D: The Constitution

E: The Bill of Rights

t) To avoid the need for the ... channel to connect to the FTP client, FTP ... mode should be used.

A: data, passive

B: data, indirect

C: data, reversed

D: control, passive

E: control, indirect

F: control, reversed

[20]

Question 2

Consider the SMTP protocol.

a) If SMTP server X forwards a message to SMTP server Y, which well-known port will Y typically use? 25 (nothing is being submitted). (1) 1

b) Suppose a client wants to send an email to u@xx.co.za. What message will be sent to the SMTP server to instruct the server to do so? MAIL TO: u@ xx.co.za (1)

c) An email to be transferred via SMTP will usually consist of a header and a body, as specified in RFC822. What separates the header from the body? blank line (1)

d) The data message is used to indicate that the email itself would be sent next. What does the client send to the server to indicate that it has transmitted the entire message? Use words to describe this 'token' that signifies the end of the message. (1)
Full stop on a line of its own

e) Is it possible to send an email to someone where the email includes a full stop on a line of its own? (Yes/No). Yes. (1)

[5]

Question 3

Consider the Telnet protocol.

a) What is the intended purpose of Telnet? Provide your answer as a descriptive phrase, rather than stating that it enables one to execute commands on remote computers. (1) 1
Virtual Terminal

b) Which well-known port is typically used by a Telnet server? 23 (1) 1

c) Suppose you are using a Telnet server. The outputs of your commands are displayed correctly. However, whenever you type something at the client, you cannot see what you are typing. Which property should you adjust at the client to rectify the problem? (1)
local echo

d) Suppose you want to use the Telnet client to interact with www.example.com using manually entered HTTP commands. Provide the full command you will enter on the command-line interface to open the connection. telnet www.example.com 80 (1)

e) When opening a Telnet connection, the Telnet client displays a character combination that one may enter to open the Telnet console. (The is often ^].) What is this sequence called by the client? escape character (1)

[5]

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

Question 4

The data link layer is used to logically connect two neighbouring nodes in a network to one another.

a) Name the three primary functions of the data link layer. Contention control / Error control / Data delineation (3) 3

b) Name one example of a data link layer protocol. Ethernet (1) 1

c) Network layers are supposed to be independent, but this is often not possible. Provide an example of how layer 1 may impact on one of the three functions you named at question a. Noisy medium requiring error correction on layer 2, vs good medium that only needs error checking on layer 2. (1) 1

[5]

Question 5

no need to repeat

On the answer sheet you will find a partial zone file for xx.co.za. You are expected to add resource records to it to enable the server to provide the answers described below. (Note that you will not repeat xx.co.za in any of your answers, because it is known that this zone file deals wit xx.co.za.)

a) The IP address of www.xx.co.za is 10.9.8.7. www IN A 10.9.8.7 (1)

b) Any traffic addressed to yy.xx.co.za should be sent to xx.org.za. (1)
yy IN CNAME xx.org.za

c) Mail to any mailbox in xx.co.za should preferably be sent to 192.168.1.1. (1)
@ IN MX <num 1> 192.168.1.1

d) If 192.168.1.1 is down, mail to any mailbox in xx.co.za should be sent to 192.168.99.99.
@ IN MX <num 2> 192.168.99.99 (1)

e) 172.16.16.16 is a name server for xx.co.za (1)
@ IN NS 176.16.16.16

[5]

TOTAL [40]

MX, 5 → priority (lower number indicates preference).

END OF PAPER

CNAME → alias/synonym
TXT → read text
HOST
NS (name server)

Q1 [11]
Q2
Q3
Q4 [5]
Q5 [2]

University of Pretoria

SEMESTER TEST: 15 MAY 2023

COURSE: Computer Science
PAPER: COS332

TIME: 90 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 10 PAGES.

ANSWER ALL QUESTIONS.

CASIO FX-82 (OR EQUIVALENT) CALCULATORS PERMITTED, AS WELL AS BASIC 5-FUNCTION CALCULATORS.
USE YOUR TEST BOOK FOR ROUGH WORK; NOTE THAT IT WILL NOT BE MARKED, BUT SHOULD BE HANDED IN.
YOU ARE ALLOWED TO WRITE ON THIS TEST PAPER.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) The following is an example of an IGP.

A: BGP
B: OSPF
C: Dijkstra
D: More than one of the above
E: All of the above

b) The ASCII standard standardises a(n) … character code.

A: 7-bit
B: 8-bit
C: 9-bit
D: More than one of the above
E: All of the above

COURSE: Computer Science
PAPER: COS332

c) Consider the following protocol negotiation string in an HTTP request:
`Accept-Charset: iso-8859-1, utf-8, utf-16, *;q=0.1`
Suppose the server supports `iso-8859-1`, `utf-8` and `Shift_JIS`. The response may then be encoded using

A: `iso-8859-1`
B: `utf-8`
C: `Shift_JIS`
D: Any of the encodings listed as options above.
E: One of two of the encodings listed as options above.

d) A BER value in ASN.1 is, in principle, encoded as a triple consisting of

A: A type, a subtype and a value.
B: A type, a length and a value.
C: A constructor and two operands.
D: A variable name, as well as its minimum and maximum values.
E: The same value encoded in binary, text and hexadecimal.

e) Routers using OSPF broadcast their … and use the received information to apply the … algorithm.

A: Routing tables; Dijkstra
B: Routing tables; Bellman-Ford
C: Routing tables; RIP
D: Version of the network topology; Dijkstra
E: Version of the network topology; Bellman-Ford

f) Which of the following is *not* a valid MIME type?

A: binary
B: example
C: image
D: message
E: model

g) Morse code uses a version of a Huffman encoding where moving along the left branch in the (balanced) tree that is constructed, is deemed to be represented by a dot, while moving to the right is deemed to be represented by a dash. The letter N is deemed to be the fifth most common character by Morse code. Hence N is represented as follows:

A: ••
B: • −
C: − •
D: − −
E: • • −

h) The session layer is layer . . . of the ISO OSI protocol stack.

A: 1
B: 2
C: 3
D: 4
E: 5

i) The transport layer protocol that is predicted to eventually replace TCP is

A: TP0
B: IP
C: TCPv2
D: QUIC
E: ARPA

j) UDP uses the following flow-control mechanism:

A: Stop and Wait
B: Sliding window
C: Unrestricted
D: More than one of the above
E: All of the above

k) Say a damaged packet arrives at its destination. The recipient may, for example, determine that the packet is damaged when the checksum does mot match. In many protocols it may then send a NAK (negative acknowledgement) to the original sender to request retransmission. Suppose A sends a TCP segment to B, but it arrives damaged at B. How is this resolved?

A: B sets the NAK TCP flag and sends it as a response to A.
B: B discards the segment and does not acknowledge it; the retransmission timer at A expires and the packet is sent again.
C: B requests the network layer at B to deal with the error, and this triggers a retransmission of the segment.
D: B corrects the damaged segment and (eventually) acknowledges receipt.
E: The lower layer protocols perform error checking, so a damaged segment will never arrive; errors will always be corrected on a lower layer.

l) A sends a TCP segment to node B. The acknowledgement field contains the value 150. The ACK flag is not set. This means

A: B may assume that bytes up to byte 149 that it had sent are acknowledged.
B: B should disregard the value 150 — it has no meaning.
C: This is a negative acknowledgement of the segment B has sent that had the sequence number 150.
D: The TCP layer at A is misconfigured; it should have set the ACK flag.
E: More than one of the above

m) Node A sends a TCP segment to node B with its ACK flag set. This may affect (or create) relevant timers at A and/or B.

What happens to one or more of the retransmission timers at A?

A: A timer is created.
B: An existing timer is set.
C: An existing timer is reset.
D: An existing timer is deleted.
E: No retransmission timer is affected.

n) Node A sends a TCP segment to node B with its ACK flag set. This may affect (or create) relevant timers at A and/or B.

What happens to one or more of the acknowledgement timers at A?

A: A timer is created.
B: An existing timer may be set.
C: An existing timer may be reset.
D: An existing timer may be deleted.
E: No acknowledgement timer will be affected.

o) Node A sends a TCP segment to node B with its ACK flag set. This may affect (or create) relevant timers at A and/or B. B receives this segment without any error.

What happens to one or more of the acknowledgement timers at B?

A: A timer is created.
B: An existing timer may be set.
C: An existing timer may be reset.
D: An existing timer may be deleted.
E: No acknowledgement timer will be affected.

p) Suppose node A is connected to node B via TCP. When the persistence timer at A expires it implies that

A: B has not yet acknowledged the last segment A sent to B.
B: A is waiting for space to free up in B's window before further data can be sent.
C: A has no data to send, so it transmits a single byte to indicate that it is still present.
D: More than one of the above
E: All of the above

q) A TCP connection in the `TIME-WAIT` state means that

A: The node is waiting for data from the node that it is connected to.
B: The connection is about to time-out if no further data is transmitted soon.
C: The connection will soon be established (as soon as the handshake is completed).
D: The node is waiting for 'lost' traffic to arrive at the port that is no longer in use, before it will be available for reuse.
E: The network is congested.

r) You see the following line as part of the `netstat` program's output on Linux:
`127.0.0.1:27117 127.0.0.1:48956 ESTABLISHED keepalive`
This means that this connection

A: Is in use, but idle.
B: Is being torn down, and will soon be closed.
C: Is busy carrying data.
D: Has been reserved for future use.
E: Is busy with a handshake.

s) A TCP node that is in the `LISTEN` state

A: Is acting as a server and waiting for a SYN message.
B: Is acting as a client and has sent a SYN message.
C: May act as a client after sending a SYN message.
D: More than one of the above
E: None of the above.

t) Which well-known port is used for control messages by an FTP server?

A: 20
B: 21
C: 22
D: 23
E: 24

Telnet
SSH

Class A address

u) What is the address (in CIDR notation) of the netblock 41.32.0.0 to 41.39.255.255?

more than 8 bits

A: 41.32.0.0/10

B: 41.32.0.0/11

C: 41.32.0.0/12

D: 41.32.0.0/13

E: None of the above

v) The private class B IPv4 addresses are:

A: 172.1.0.0–172.1.255.255

B: 172.2.0.0–172.3.255.255

C: 172.4.0.0–172.7.255.255

D: 172.8.0.0–172.15.255.255

E: 172.16.0.0–172.31.255.255

w) The following type of router connects an organisation's network to other networks.

A: Ingress routers

B: Egress routers

C: Boundary routers

D: More than one of the above

E: All of the above

x) Dynamic IP addresses are often assigned to subscribers by their ISPs. In some cases the ISP offers a static address as an additional service, so that servers hosted by the subscriber are easy to reach. In some cases such a static address is assigned through network address translation. This means that the static IP address assigned by the ISP

A: Is the public address assigned to the subscriber's connection.

B: Is a public address that is redirected to the IP address associated with the subscriber's connection.

C: Is the private address assigned to the subscriber's connection.

D: Is a private address that is redirected to the IP address associated with the subscriber's connection.

E: Is a dynamic address linked to an FQDN that uses dynamic DNS.

y) If one sends a message to an IP multicast address, the message will be delivered to

A: The node to which that IP address has been assigned.

B: All nodes on the local network.

C: All nodes that are members of some group.

D: The network itself, rather than to a host on the network.

E: To the network gateway that connects the network to other networks. **[25]**

Question 2

Your organisation uses the following IP address as part of the block assigned to them: 170.170.170.170/16.

They decide to subnet their address space such that they have at least 1000 subnets that can each handle at least 60 hosts.

a) What netmask will they use for the subnets? 255.255.255.~~0~~ 192 (1)

b) Consider *subnet* number 170. What is the network address of this subnet? Express it using CIDR notation. 170.170.~~36~~.~~0~~ /26 (2)

c) What is the broadcast address of subnet 170? 170.170.~~255~~.255 42 128 191 (1)

63 d) What is the last (or 'biggest') address that may be assigned to a host on subnet 170? 42 170.170.42.190 (1)

86 e) On which subnet will the host with the IP address 170.170.170.170 be? Just provide the number of the subnet. 682 (1)

The organisation decides that it needs a bigger subnet in the laboratory where 170.170.170.170 is located. They decide to combine a number of subnets into a supernet that can handle at least 1000 hosts. However, they want to sacrifice as few subnets as possible to create this supernet.

f) Provide the netmask associated with this supernet. 255.255.~~191~~.0 252 (1)

g) Provide the network address of this supernet using CIDR notation. 170.170.~~191~~.0/22 168 (2)

h) What is the *number* of the first subnet that will be lost once this supernet has been created? 672 (1)

[10]

Question 3

a) Consider the following UTF-8 byte sequence:
`E1 A1 80 41 EF A4 9A`

U+1840
U+0041
U+F919

Convert this sequence to Unicode codepoints in the form U+xxxx.

Note that the answer sheet contains more open blocks than what you need. Just place a dash in blocks that are not necessary.

b) Consider the following Unicode codepoints:
`U+00C6 U+10000`

C3 86
F0 90 80 80

Convert this sequence to a UTF-8 byte sequence where each byte is represented in hexadecimal.

Note that the answer sheet contains more open blocks than what you need. Just place a dash in blocks that are not necessary.

[5]

Question 4

The TCP header includes a flags field that consist of various flags. Name any five (5) *other* fields that occur in the TCP header. **[5]**

Source port
Destincution port
Check sum
Window
Sequence number.

5

Question 5

Node A is busy communicating with node B via a TCP connection. At some time t the next byte that A will send is byte number 300. The next byte that B will send is byte number 500. The window size at A is 1000. The window size at B is 100. At time t, both A and B have successfully received all previous messages sent by its counterpart.

At time $t + 1$ A wants to send 200 bytes to node B. Call the message that A actually sends m_1.

At time $t + 2$ B consumes 100 bytes from its buffer.

At time $t + 3$ B sends a message containing 50 bytes to A. Let us call this message m_3.

At time $t + 4$ A sends a message to B to acknowledge receipt of m_3. A also includes all residual data it may have to send. Let us call this message m_4.

At time $t + 5$ B sends an empty acknowledgement message to A to acknowledge m_4. Let us call this acknowledgement message m_5.

a) What is the sequence number included with m_4? 500 ✗ 400 **(1)**

b) What is the acknowledgement number included with m_4? 550 ✓ **(1)**

c) What is the length of m_4? 0. 100 **(1)**

d) What is the window advertisement included with m_4? 1000 ✗ **(1)**

e) What is the window advertisement included with m_5? 0. ✓ **(1)**

[5]

2

TOTAL **[50]**

END OF PAPER

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Engineering, Built Environment and IT

Department of Computer Science

Module name

COS 332

Assessment opportunity

21 July 2023

Examiners

Internal: Prof MS Olivier
External: Prof M Coetzee (NWU)

Instructions

1. Read the question paper carefully and answer all the questions below.
2. The assessment opportunity comprises of 8 questions on **21** pages.
3. An answer sheet and script for calculations are provided separately.
4. You have **2.5 hours** to complete the paper.
5. This is a closed book paper. You are therefore not allowed to have any study material with you.
6. You may use a basic calculator (+, -, ×, ÷, √, =) or any variant of the Casio FX-82 calculator (or equivalent).
7. Answer all questions.
8. All answer sheets must be completed in indelible ink. Answer sheets completed in pencil or erasable ink will not be marked and you will not qualify for an additional assessment opportunity.
9. When a question can be answered by means of an acronym it is not necessary to provide the phrase that the acronym represents.
10. Please switch off your cell phone, and keep it off for the duration of the paper.
11. Students are allowed to write on the examination paper.
12. Please consult the module website about the perusal.

Integrity statement:
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1 1) A layer in the protocol stack generally provides services for

A: The layer above it ✓ 7
B: The layer below it services
C: Its peer layer
D: The application layer 1
E: The physical layer

2) The nom.za second-level domain is intended to allow ... to register subdomains within it.

A: Nomads
B: Individuals
C: Parties who handle nominations for potential office bearers
D: Those who provide authoritative definitions of nomenclature
E: None, because no such second-level domain exists

3) The primary metric used by interior gateway protocols is

A: Cost
B: Hop count
C: Distance
D: Static routing tables
E: Speed
F: Noise

4) The class of protocols used for routing between autonomous systems is known as

A: IGPs
B: BGPs
C: EGPs
D: RIP
E: Peering

5) Which of the following data streams generally work(s) better with an unreliable layer 4 protocol?

A: Web page
B: File transfer
C: Audio
D: More than one of the above
E: All of the above

6) If MIME is deemed to be a data communications protocol it would fit best on layer ... of the ISO OSI model.

A: 7
B: 6
C: 5
D: 4
E: 3

7) When writing an application that will serve as a server in a client-server architecture, the application would (in its role as server) open a socket that would

A: Be specified as an address and port number
B: Initiate connections (only client initiates connection)
C: Wait for connections
D: More than one of the above
E: All of the above

1 8) Which of the following characters is not present in standard ASCII?

A: A
B: &
C: #
D: £
E: ~

1 9) Which ISO layer enables process-to-process communication via the network?

A: 6
B: 5
C: 4
D: 3
E: 2

7 Application
6 Presentation
5 Session
4 Transport (pipe)
3 Network
2 Data link
1.

0 10) The OSI layer that, amongst others, attempts to emulate database transactions where messages can be 'rolled back' if the final message in a sequence is not sent, is layer

A: 2
B: 3
C: 4
D: 5
E: 6

0 11) The capetown. sTLD has to adhere to policies prescribed by

A: ICANN
B: ZADNA
C: ISO
D: More than one of the above
E: All of the above

12) The SMTP submit port is

1

A: 25
B: 100
C: 443
D: 587
E: None of the above

13) With which architecture was the *Napster* service associated?

1

A: Client-server
B: Peer-to-peer
C: 3-tier
D: 4-tier
E: 5-tier

14) Which condition code will be returned by an SMTP server after successfully handling a `helo` or `ehlo` message?

1

A: 0
B: 150
C: 250
D: 350
E: 450

15) To avoid the need for the ... channel to connect to the FTP client, FTP ... mode should be used.

0

A: data, passive
B: data, indirect
C: data, reversed
D: control, passive
E: control, indirect
F: control, reversed

ingress gateway protocol.

16) The following is an example of an IGP.

0

A: BGP
B: OSPF
C: Dijkstra
D: More than one of the above
E: All of the above

17) A BER value in ASN.1 is, in principle, encoded as a triple consisting of

0

A: A type, a subtype and a value.
B: A type, a length and a value.
C: A constructor and two operands.
D: A variable name, as well as its minimum and maximum values.
E: The same value encoded in binary, text and hexadecimal.

18) Routers using OSPF broadcast their ... and use the received information to apply the ... algorithm.

0

A: Routing tables; Dijkstra
B: Routing tables; Bellman-Ford
C: Routing tables; RIP
D: Version of the network topology; Dijkstra
E: Version of the network topology; Bellman-Ford

19) The transport layer protocol that is predicted to eventually replace TCP is

1

A: TP0
B: IP
C: TCPv2
D: QUIC
E: ARPA

1

20) Morse code uses a version of a Huffman encoding where moving along the left branch in the (balanced) tree that is constructed, is deemed to be represented by a dot, while moving to the right is deemed to be represented by a dash. The letter N is deemed to be the fifth most common character by Morse code. Hence N is represented as follows:

A: ••
B: • –
C: – •
D: – –
E: • • –

0

21) A sends a TCP segment to node B. The acknowledgement field contains the value 150. The ACK flag is not set. This means

ACK = 150
A B
Flag is not set.

A: B may assume that bytes up to byte 149 that it had sent are acknowledged.
B: B should disregard the value 150 — it has no meaning.
C: This is a negative acknowledgement of the segment B has sent that had the sequence number 150.
D: The TCP layer at A is misconfigured; it should have set the ACK flag.
E: More than one of the above

0

22) Node A sends a TCP segment to node B with its ACK flag set. This may affect (or create) relevant timers at A and/or B.
What happens to one or more of the acknowledgement timers at A?

A: A timer is created.
B: An existing timer may be set.
C: An existing timer may be reset.
D: An existing timer may be deleted.
E: No acknowledgement timer will be affected.

0

23) Suppose node A is connected to node B via TCP. When the persistence timer at A expires it implies that

A: B has not yet acknowledged the last segment A sent to B.
B: A is waiting for space to free up in B's window before further data can be sent.
C: A has no data to send, so it transmits a single byte to indicate that it is still present.
D: More than one of the above
E: All of the above

1

24) A TCP connection in the `TIME-WAIT` state means that

A: The node is waiting for data from the node that it is connected to.
B: The connection is about to time-out if no further data is transmitted soon.
C: The connection will soon be established (as soon as the handshake is completed).
D: The node is waiting for 'lost' traffic to arrive at the port that is no longer in use, before it will be available for reuse.
E: The network is congested.

0

25) A TCP node that is in the `LISTEN` state

A: Is acting as a server and waiting for a SYN message.
B: Is acting as a client and has sent a SYN message.
C: May act as a client after sending a SYN message.
D: More than one of the above
E: None of the above.

1

26) What is the address (in CIDR notation) of the netblock 41.32.0.0 to 41.39.255.255?

A: 41.32.0.0/10
B: 41.32.0.0/11
C: 41.32.0.0/12
D: 41.32.0.0/13
E: None of the above

~~41.00100000.0.0~~

~~41.00100Φ10.0.0~~

27) The private class B IPv4 addresses are:

1

A: 172.1.0.0–172.1.255.255
B: 172.2.0.0–172.3.255.255
C: 172.4.0.0–172.7.255.255
D: 172.8.0.0–172.15.255.255
E: 172.16.0.0–172.31.255.255

28) The following type of router connects an organisation's network to other networks.

0

A: Ingress routers
B: Egress routers
C: Boundary routers
D: More than one of the above
E: All of the above

29) If one sends a message to an IP multicast address, the message will be delivered to

0

A: The node to which that IP address has been assigned.
B: All nodes on the local network.
C: All nodes that are members of some group.
D: The network itself, rather than to a host on the network.
E: To the network gateway that connects the network to other networks.

30) The syntactic structure of an ASN.1 message is defined using

A: A grammar
B: An informal description in a natural language
C: A diagrammatic depiction of the message
D: A bit pattern
E: A list of the types of the components of the message

31) Suppose a message consists of JPG images embedded in HTML text. Which MIME type and subtype will be used to describe the message?

A: `text/plain`
B: `text/html`
C: `image/jpeg`
D: `multipart/mixed`
E: `multipart/alternative`

32) Suppose A and B are communicating via TCP. A has n bytes to transmit. As far as A is aware, the window size at B is $m < n$. This means that

not enough space in buffer

A: A cannot transmit.
B: A can transmit m bytes.
C: A can transmit n bytes.
D: More than one of the above
E: All of the above

33) An IP datagram addressed to 10.10.1.2 arrives at node R. R is not directly connected to the destination address. The routing table at R includes the following entries:

```
10.10.10.0/24 192.168.1.1 eth1   10.10.0.0/16 192.168.2.2 eth2
10.0.0.0/8    192.168.3.3 eth3   0.0.0.0/0    192.168.4.4 eth4
```

To which next hop does R forward the datagram?

A: 192.168.1.1
B: 192.168.2.2
C: 192.168.3.3
D: 192.168.4.4
E: More than one of the above

34) Which of the following IPv6 addresses correspond with the IPv4 address 10.10.10.10?

A: 0::A:A:A:A

B: 0::AA:AA

C: 0::FFFF:A:A:A:A

D: 0::FFFF:AA:AA

E: 0::FFFF:A0A:A0A

35) Consider the IPv6 address:
`fd12:3456:789a:bcde:f0fe:dcba:9876:5432`
Which of the following is true about the part `12:3456:789a`?

A: It was supplied by an ISP/RIR.

B: It designates a subnet.

C: It is a random number.

D: It was calculated from the MAC address of the interface.

E: None of the above

36) Consider the IPv6 address:
`fd12:3456:789a:bcde:f0fe:dcba:9876:5432`
Which of the following is true about the part `bcde`?

A: It was supplied by an ISP/RIR.

B: It designates a subnet.

C: It is a random number.

D: It was calculated from the MAC address of the interface.

E: None of the above

37) Consider the IPv6 address
`fe80:1234:5678:9abc:def0:fedb:a987:6543/128`
This address

A: Is an SLA address.

B: Is a link-local address.

C: Is a site-local address.

D: Is a global address.

E: Would currently not be a legal address.

38) Which of the following prefixes indicate an IPv4-mapped address expressed in IPv6 notation?

A: ::ffff::/~~80~~ 96

B: ::ffff:0:0/~~80~~ 96

C: 0:0:0:0:ffff::/~~80~~ 96

D: More than one of the above

E: All of the above

time to live.

39) Suppose a router receives an IP datagram *d* and its TTL become 0. What may the router do next?

A: Discard *d*.

B: Forward *d* to the next hop.

C: Send a time exceeded ICMP message to the source.

D: More than one of the above

E: All of the above

IPs → layer 3.

40) An IP datagram may be fragmented to

A: Expedite routing.
B: Fit IP datagrams in size-constrained layer 2 frames.
C: Ensure that fixed size packets are sent to the lower layer.
D: Fit IP datagrams in size-constrained layer 4 segments.
E: Minimise data loss if a packet is discarded.

41) What is the name of the file that contains IP addresses and corresponding names that may be used for name resolution on almost every computer connected to the Internet?

A: `names.txt`
B: `resolv.conf`
C: `names`
D: `resolv`
E: `hosts`

42) IEEE 802.3 defines

A: Ethernet
B: LLC
C: HDLC
D: Token ring
E: More than one of the above

43) The Ethernet address `FF:FF:FF:FF:FF:FF` is

A: A MAC address of a specific node
B: A broadcast address
C: A group address
D: Invalid
E: None of the above

44) The Ethernet card with the MAC address `00:50:DA:45:B5:E8` was manufactured by 3Com. Which of the following cards was/were therefore also manufactured by 3Com?

A: `00:C0:DF:06:47:72`
B: `00:50:DA:06:47:72`
C: `00:C0:DF:45:B5:E8`
D: More than one of the above
E: All of the above

token ring
bully algorithm

45) When no monitor's presence is detected on an 802.5 network

A: The hub appoints a new monitor.
B: The previous monitor resumes its duties.
C: The station with the highest priority starts acting as monitor.
D: A new monitor is elected by the remaining nodes.
E: The network simply continues operating because it does not really need a monitor.

46) An Ethernet card that is in promiscuous mode will

A: Occasionally send obscene messages.
B: Continue to transmit even after a collision has occurred.
C: Accept messages not addressed to it.
D: Not forward the token to another node once it has received it.
E: Flood the network with traffic.

47) Manchester encoding is better than NRZ because

A: Manchester is self-synchronising.
B: The baud rate of Manchester is higher than that of NRZ.
C: The baud rate of Manchester is lower than that of NRZ.
D: The wires may be switched in the case of Manchester.
E: None of the above

48) Given that the speed of light is $3 \times 10^8 ms^{-1}$, how long does it take a signal to travel from an earth station to a geostationary satellite and back to another earth station?

A: 0,12s
B: 0,24s
C: 0,36s
D: 0,48s

49) Advantages of a LEO satellite for data communications include the following:

A: Since it is placed closer to earth than geosynchronous satellites, a smaller transmitter can be used from earth.
B: It orbits around the earth.
C: It is not affected as much as geosynchronous satellites by mountains and other natural obstacles.
D: More than one of the above
E: All of the above

50) A master-slave protocol may solve the following problem on layer 2:

A: Media access control
B: Error control
C: Routing
D: Polling
E: None of the above

[50]

Question 2

Assume you receive the following 8 bit characters encoded with a single bit correction Hamming code. Bits are numbered from right to left and even parity is used. In each case say which bit has changed (in the range 1–12). If no bit has changed, simply write 0 on your answer sheet.

a) `000000000110` (2)

b) `101000000110` (2)

c) `011100110010` (2)

Also answer the following questions about Hamming codes.

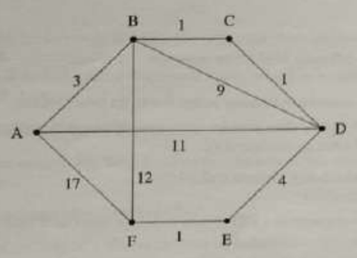
d) How many parity bits need to be added to create a code that would provide single bit correction for 12 data bits? (2)

e) Consider a single-bit Hamming code capable of correcting single bit error that results in 20-bit messages. How many of those 20 bits would be data bits? (2)

[10]

✓

Question 3



In the following table Dijkstra's algorithm has been used to calculate the cheapest routes from node A. Complete the table by writing the values that have been omitted (and indicated with lowercase letters in brackets) on your answer sheet. Question marks also indicate values that have been omitted but you do not have to supply those values.

				Cost					Prior				
Step	S	W	X	B	C	D	E	F	B	C	D	E	F
1	A	?	B	3	∞	11	∞	13 (a)	A	–	A	–	?
2	A, B	?	C	3	4	11	∞	15	A	B	A	–	B
3	?	?	?	3	4	(b) 5	(c)	15	A	B	?	–	B
4	?	?	?	3	4	?	9	15	A	B	?	D	B
5	?	?	?	3	4	?	9	(d)	A	B	C	D	(e)

[5]

✓✓✓ ✓
A B D FC E

✓

Question 4

Suppose a network uses the Bellman-Ford algorithm to perform routing. At some time t_0 the routing tables at routers C, D and E contain the information provided below. Each routing table consists of three columns: destination, cost, and the next hop to achieve delivery at the indicated cost.

3 jump   4 jump   dest cost next

Node C		
A	9	B
B	2	B
D	3	D

Node D		
A	20	E
C	3	C
E	4	E
G	8	E

Node E		
A	6	F
B	2	F
D	4	D
F	5	F

The only exchanges of routing tables that occur during the time period of interest are the following: At time $t_1 > t_0$ node C sends its routing table to node D. At time $t_2 > t_1$ node E sends its routing table to node D. C → D then E → D

a) What will the entry for destination A be in node D's routing table, after receiving (and processing) the routing table from node C (but prior to t_2)? (1)

b) What will the entry for destination B be in node D's routing table, after receiving (and processing) the routing table from node C (but prior to t_2)? (1)

c) What will the entry for destination C be in node D's routing table, after receiving (and processing) the routing table from node C (but prior to t_2)? (1)

d) What will the entry for destination A be in node D's routing table, after receiving (and processing) the routing table from node E? (1)

e) What will the entry for destination B be in node D's routing table, after receiving (and processing) the routing table from node E? (1)

[5]

Question 5

Consider the following zone files at various DNS servers. Assume that records and fields that are not shown (such as SOA records) are not material.

root.

2		
a	A	11
b	A	7
a	NS	4
b	NS	6

3		
a	A	13
b	A	10
a	NS	2
b	NS	5

4		
a	A	14
b	A	15
a	NS	2
b	NS	5

5		
a	A	3
b	A	5
a	NS	6
b	NS	1

9		
a	A	4
b	A	3
a	NS	3
b	NS	4

10		
a	A	8
b	A	12
a	NS	3
b	NS	5

Note that these files break some of the rules for zone files; the root name server, for example, contains A resource record. A number of entries create cycles. However, these issues are not material for the questions that follow. The number in bold at the top of each table is the address of the node on which that zone file occurs.

Node 9 is the root name server.

Perform recursive name resolutions for the following FQDNs, for the query provided. If the FQDN is preceded by an NS, you should perform an NS query. If the FQDN is preceded by an A, you should perform an A query. Write the address found (as a single number), or write N/A if the query cannot be resolved wit the information provided.

a) NS `b.a` (1)
b) A `b.a` (1)
c) NS `a.a.a` (1)
d) A `a.a.a` (1)
e) NS `a.b` (1)
f) A `b.a.b` (1)

Now insert resource records in the appropriate zone files to achieve the goals set out in the questions below. If you, for example, want to insert an NS record of the form `c NS 7` at node 5, simply write "`At 5: c NS 7`".

g) Mail to `b.a` should be delivered to address 10. You are *not* allowed to add a record to node 3, 6 or 9 to achieve this. MX < > (1)
h) If node 10 is unavailable, mail to `b.a` should be delivered to address 15. You are *not* allowed to add a record to node 3, 6 or 9 to achieve this. MX, < > (1)
i) `c.a.a` should be an alias for `b.a.b`. CNAME (1)
j) The IPv6 address of `a.a.b` is 2001::2001. C NAT (1)

[10]

✓Question 6

a) Consider the following UTF-8 byte sequence:
`D0 A3 D0 BA D1 80`

Convert this sequence to Unicode codepoints in the form U+xxxx.

Note that the answer sheet contains more open blocks than what you need. Just place a dash in blocks that are not necessary.

b) Consider the following Unicode codepoints:
`U+1F64F U+00A1`

Convert this sequence to a UTF-8 byte sequence where each byte is represented in hexadecimal.

Note that the answer sheet contains more open blocks than what you need. Just place a dash in blocks that are not necessary.

[5]

✓Question 7

Suppose nodes A an B want to communicate using TCP. Three messages are sent to establish the connection; refer to these messages as m_1, m_2 and m_3. Assume sequence numbers start at 0.

a) Name a flag that will be set in the header of m_1. (1)
b) Name two flags that will be set in the header of m_2. (1)
c) What is the acknowledgement value of m_1? (1)
d) What is the acknowledgement value of m_2? (1)
e) What is the acknowledgement value of m_3? (1)

[5]

Question 8

Your organisation uses the following IP address as part of the block assigned to them:
204.204.204.204/16.

They decide to subnet their address space such that they have at least 2000 subnets that can each handle at least 30 hosts.

a) What netmask will they use for the subnets? (1)

b) Consider *subnet* number 204. What is the network address of this subnet? Express it using CIDR notation. (2)

c) What is the broadcast address of subnet 204? (1)

d) What is the last (or 'biggest') address that may be assigned to a host on subnet 85? (1)

e) On which subnet will the host with the IP address 204.204.204.204 be? Just provide the number of the subnet. (1)

The organisation decides that it needs a *much* bigger subnet in the laboratory where 204.204.204.204 is located. They decide to combine a number of subnets into a supernet that can handle at least 4000 hosts. However, they want to sacrifice as few subnets as possible to create this supernet.

f) Provide the netmask associated with this supernet. (1)

g) Provide the network address of this supernet using CIDR notation. (2)

h) When using this supernet, it is possible to address more hosts than would have been possible using the subnets that have been sacrificed to create the supernet. How many more? (1)

[10]

[100]

TOTAL

END OF PAPER